Routine Edits, Measured:

A Paired Evaluation of Perceptual-Hash Interdiction Under Non-Adversarial Transformations

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Companion to Bravo and Castanon Acuna, Weapons of Mass Deception (SSRN 7222219, 2026), which carries the governance argument this note measures one corner of.

Abstract

Hash-based interdiction is the deployed defence against the redistribution of intimate imagery: a person enrols a perceptual hash of an image they hold, and participating platforms block uploads whose hash falls within a threshold of it. The primitive's behaviour under ordinary, non-adversarial edits determines how much of that promise survives contact with everyday redistribution, and the public record on it is thin, aggregated, and measured mostly on hash functions that nobody deploys. We report a paired measurement on 5,000 everyday photographs (COCO val2017) of three hashes: Meta's reference PDQ, the deployed open standard; the 64-bit DCT pHash of the academic literature; and a 256-bit block-mean hash that is not a DCT descendant. Seven transformation families at four intensities each (JPEG recompression, resize round-trip, Gaussian blur, additive noise, brightness, center crop, small-angle rotation) are applied to every image, every hash scores the same edited copy, and we report the *residual evasion rate*, the share of edited images no longer matched to their own original at the deployed threshold, with Wilson intervals, McNemar tests on identical pairs, image-level bootstrap intervals, and the sample size each effect would require. Non-geometric edits are practically invisible to all three hashes: residual rates of 0.0% to 0.1% at every intensity, including JPEG quality 30 and blur of radius 3.5. Geometric edits dominate: PDQ fails to match 79.9% of center-cropped copies (19.8% when 95% of the area is kept, 99.8% at 85%) and 97.2% of rotated ones (88.8% at two degrees, 100% from five). pHash fails on 59.0% and 72.6%, the block-mean hash on 53.2% and 64.6%; the non-DCT hash's advantage over PDQ is 26.7 and 32.6 percentage points (bootstrap intervals exclude zero; McNemar χ^2 5,310 and 6,504). On the non-geometric families, differences of 0.1 percentage points reach $p < 0.01$ at $n = 20{,}000$, which is the usual large-sample caution. Only 9 of 5,000 photographs fall under PDQ's quality floor, against 318 of 600 in an earlier procedurally generated corpus whose numbers we withdraw. We discuss what the result means for deployed interdiction, for governance after the 2026 white-box analysis of PhotoDNA and the generative inversion attacks on PDQ, and for the fully synthetic case, which the open-weight image and video generators released between late October and early November 2026 move further out of reach of any hash: a generated image has no source to enrol, and a model that runs offline carries neither the operator's filter nor a durable mark. All code, committed aggregate data and figures regenerate from the repository; no abusive material was held or processed.

Keywords: perceptual hashing; PDQ; pHash; robustness; non-consensual intimate imagery; content moderation; trust and safety; reproducibility; open-weight generative models.

1. Introduction

Three services define hash-based interdiction as it exists in 2026. StopNCII.org (SWGfL and the Revenge Porn Helpline) lets an adult generate a perceptual hash of an intimate image on their own device and shares the hash, never the image, with participating platforms. Take It Down (NCMEC) does the same for imagery of a person who was under eighteen when it was taken. Since 19 May 2026 the United States' TAKE IT DOWN Act obliges covered platforms to remove reported non-consensual intimate imagery, including known identical copies, within 48 hours, with civil penalties enforced by the Federal Trade Commission (Bravo and Castanon Acuna, 2026, section 2.4). Every one of these systems rests on one primitive: a perceptual hash, compared under a Hamming-distance threshold, is expected to recognise an image after the ordinary transformations redistribution inflicts on it.

How well it does so is a measurable property, and the measurement is owed. The published evaluations either aggregate transformations into a single robustness score (McKeown and Buchanan, 2023), or measure hashes that are not the one platforms deploy, or come from the designers of the hash. Meta's PDQ is the open standard the ecosystem converged on, with its own reference implementation and a documented default threshold of 31 bits out of 256; it is the hash Take It Down uses (Hawkes et al., 2025). The question this note answers is narrow and, we think, prior to every other question about hash-based interdiction: when an ordinary person recompresses, resizes, crops, rotates, brightens or blurs a photograph in the way phones, messaging apps and screenshots do every day, what share of the copies does the deployed hash stop recognising, and does a hash of a different family fail in the same places?

We answer it with a paired protocol. Each of 5,000 photographs is edited under seven transformation families at four intensities each; every hash scores the *same* edited copy; and we report the share of copies no longer matched to their own original, with the statistics a reviewer asks for when two methods are compared on identical inputs. The protocol, the statistics module and the figure renderer are committed code; the aggregate numbers behind every table and figure are committed data; and the corpus is a public photographic benchmark that never enters the repository. Nothing in the measurement touches abusive material, because robustness under a transformation is a property of the hash function and the transformation, not of the content of the image.

Terminology. We use *residual evasion rate* for the share of transformed images that a hash-matching system fails to match to their own original at its operating threshold. We reserve *residual population*, as used in our companion working paper, for the set of people that deployed hash-based tools structurally exclude. The two are related but distinct, and conflating them would overstate what a robustness measurement can say about who is protected.

Contributions. (1) A reproducible, paired protocol for routine-transformation robustness with Wilson intervals, McNemar tests, image-level bootstrap and a sample-size table, released as code. (2) Its results on 5,000 photographs for the deployed hash (PDQ) and two alternatives, disaggregated by

family and intensity. (3) A clear separation between the two edits that defeat matching (crop and small-angle rotation) and the five that do not, with the breaking points: PDQ loses one copy in five when 5% of the area is cropped away and nine in ten at two degrees of rotation. (4) Evidence that a hash from a different mathematical family fails substantially less on exactly those edits, which is a design argument for fusing dissimilar hashes rather than stacking DCT descendants. (5) A dated, primary-sourced account of what the image and video generators released between late October and early November 2026, open-weight and hosted, from the United States and China, change for hash interdiction and for concept filters.

2. Background and Related Work

2.1 The primitive

A perceptual hash maps an image to a short bit string such that visually similar images map to nearby strings under Hamming distance (Venkatesan et al., 2000; Farid, 2021). PDQ (Meta, 2019) resizes to 64 × 64 luma through a two-pass box filter, takes a two-dimensional DCT, keeps 16 × 16 low-frequency coefficients and thresholds them at their median, giving 256 bits of which exactly 128 are set; the documented match threshold is 31 bits, and the implementation also emits a quality score, with images at or below 49 recommended for discard. The 64-bit pHash of the literature (Zauner, 2010) resizes to 32 × 32, keeps an 8 × 8 DCT block and thresholds at the median. Block-mean hashing (Yang, Gu and Niu, 2006) thresholds the mean luminance of a grid of blocks and shares no transform with the DCT family, which is why it serves here as the control for "fails differently". PhotoDNA (Microsoft, 2009) is the proprietary incumbent of the shared-database ecosystem; in March 2026 Deryck, Leblanc-Albarel and Preneel published the first mathematical description of a function with identical outputs to it on a large test set, showed the function to be piece-wise linear and differentiable, and constructed exact collisions and second preimages by gradient optimisation.

2.2 Robustness evaluations

McKeown and Buchanan (2023) ran the largest public evaluation: a million-image Flickr corpus, seven hashes including PDQ, pHash and Blockhash, seven content-preserving transformations, with inter-image Hamming distributions reported in full. Their robustness results are reported per transformation but at one intensity each, and the paper's centre of gravity is the inter-image distribution and its false-positive floor; our earlier work carried that floor through to review-queue economics (Bravo and Castanon Acuna, 2026, sections 2.2 and 2.3). Steinebach, Berwanger and Liu (2022; 2023) showed that cropping and rotation are the transformations robust image hashing fails on and proposed segmentation-based hashing to address them; Steinebach (2023) analysed PhotoDNA's behaviour empirically. The protocol below adopts their framing of crop and rotation as the hard cases and measures the deployed hash against it at graded intensities, with paired statistics, on a corpus whose licence terms and composition are public.

2.3 Adversarial attacks, inversion, and defences

Adversarial evasion and collision are well studied and out of scope here: Hao et al. (2021) manipulated perceptual-hash applications with gradient-based perturbations; Jain, Creţu and de Montjoye (2022) evaluated detection-avoidance attacks against perceptual-hash client-side scanning; Struppek et al. (2022) broke Apple's NeuralHash within weeks of its release; Prokos et al. (2023) attacked PhotoDNA and PDQ with adversarial machine learning; Yang et al. (2025) proposed certified robust training (CertPHash) as a defence. Hawkes, Weinert, Almeida and Mehrnezhad (2025) showed that low-budget inversion attacks based on generative adversarial networks reconstruct recognisable pre-images from PDQ and NeuralHash values on consumer hardware, directly contradicting the "cannot be reverse engineered" assurance on Take It Down's public FAQ at the time, and recommended private set intersection for hash matching. Daskalaki, Kokolaki and Fragopoulou (2025) argue that perceptual hashes of abuse material may be personal data under European law. Our measurement is deliberately non-adversarial: it asks what happens to matching when nobody is trying to defeat it.

2.4 What this note adds

Disaggregation by intensity on the deployed hash; a paired design with the statistics that make a between-hash comparison defensible; a non-DCT control; and a corpus, code and data arrangement that lets any reader regenerate every number in 25 minutes on a laptop. None of these is a new primitive. Stating that first is a design decision.

3. Method

3.1 Corpus

We use the 5,000 validation images of COCO 2017 (Lin et al., 2014), downloaded from the public distribution (val2017.zip, about 815 MB) on 14 September 2026 and kept outside the repository. They are everyday Flickr photographs, mostly 640 pixels on the long side, with the 1/f spectral statistics and the repetition of real scenes that a hash function's behaviour depends on. The COCO annotations are CC-BY 4.0; the images carry their individual Flickr licences, a majority of them non-commercial variants. Measuring a hash function's error behaviour over the corpus and publishing the resulting rates redistributes no image and creates no derivative work, and the research arrangement is documented in the repository's compute note. We did not filter by PDQ quality: a robustness question asks whether the same image survives an edit, so every image is kept, and we report how many Meta's quality floor would have discarded (9 of 5,000). An earlier run of the same protocol on 600 procedurally generated images (coloured shapes on gradients) is kept in the repository as a reproducibility fixture only; 318 of those 600 fall under the quality floor, which is the generator telling the reader they are not photographs, and no number from that run is reported as a finding.

3.2 Hash functions and thresholds

PDQ is computed with Meta's reference implementation through the `pdqhash` Python binding (version 0.2.8), 256 bits, match threshold 31 bits, quality floor 49, exactly as documented by Meta. pHash is our implementation of the classic algorithm: greyscale, resize to 32 × 32 with Lanczos resampling,

two-dimensional DCT, top-left 8 × 8 block, median taken over the 63 non-DC coefficients, 64 bits. The third hash is a block-mean hash over a 16 × 16 grid of the Lanczos-resized greyscale image thresholded at the global median, 256 bits; we call it Blockhash-GM to distinguish it from the Commons Machinery reference blockhash, whose per-block thresholding differs and whose inter-image statistics differ in turn. Its role here is as a non-DCT control for robustness, and its numbers should not be read as those of the reference library. Thresholds for pHash and Blockhash-GM are PDQ's 31/256 scaled to the same normalised distance: 8/64 and 31/256. A match is a Hamming distance at or below the threshold between the hash of the original and the hash of its edited copy.

3.3 Transformation protocol

Seven families, four intensities each, 28 edited copies per image, all implemented with Pillow 12.3.0 and NumPy, deterministic given the image. JPEG recompression at quality 90, 70, 50 and 30 (Pillow's encoder with its default chroma subsampling). Center crop keeping 95%, 85%, 70% and 50% of the image area (side scale equal to the square root of the kept fraction), then Lanczos resize back to the original dimensions, so that only content is removed and the hash sees an image of the original size. Resize round-trip by factors 0.8, 0.5, 0.3 and 0.15 and back up, Lanczos both ways. Additive Gaussian noise with σ of 5, 10, 20 and 35 grey levels, generator seeded from the SHA-256 of the image's own pixel bytes so that the same image always receives the same noise. Rotation about the centre by 2, 5, 10 and 15 degrees, bicubic resampling, canvas size kept, uncovered corners filled black. Brightness shifts of +10, −10, +25 and −25 grey levels with clipping. Gaussian blur of radius 0.5, 1.0, 2.0 and 3.5 pixels. The set is the protocol of our September 2026 draft; the implementation fixes every parameter a reader would otherwise have to guess at (resampling kernel, corner fill, JPEG library, noise seed), and all of them are in `research/vdlab/transforms.py`.

3.4 Paired design and statistics

Each edited copy is produced once and hashed by all three functions, so every between-hash comparison is on identical image-transformation pairs. For each family and hash we report the residual evasion rate pooled over the four intensities (n = 20,000 pairs) with a Wilson score interval (Wilson, 1927) rather than the normal approximation, because rates near 0% and 100% are exactly where the normal interval fails and exactly where these results land; we also report the rate at each intensity (n = 5,000). Between hashes we report McNemar's test on the discordant pairs with continuity correction, falling back to the exact binomial when fewer than 25 pairs are discordant (McNemar, 1947); the difference in residual rate with a 95% bootstrap interval from 2,000 replicates resampled *by image*, so that the four intensities of one image travel together and the interval honours the pairing; and, for the two families with a non-trivial baseline, the number of images per condition required to detect a 5, 10, 15, 20 or 30 percentage-point reduction from PDQ's residual rate at α = 0.05 and power 0.80 by the two-proportion normal approximation, which assumes independent groups and is therefore conservative for a paired design. The statistics module is NumPy and the standard library only, with a known-answer self-test, so that every number here is reproducible from a formula the reader can inspect.

3.5 Reproducibility

One command runs the whole measurement and writes a JSON results file; a second renders the figures and tables from that file; the results file that backs the published figures is committed to the repository so that figures regenerate from committed code and committed numbers. The run of record completed on 14 September 2026 in 1,452 seconds (24 minutes) on one core of an Apple M5 laptop; it was repeated on 8 November 2026 in the environment listed in Appendix A to confirm the numbers regenerate.

```
.venv-research/bin/python research/run.py --corpus ~/datasets/val2017 --routine --limit 5000
--seed 1 --i-confirm-benign-corpus
.venv-research/bin/python research/plot_routine.py paper/data/routine-coco-val2017.json --
corpus-label "COCO val2017" \
    --svg paper/figures-en/paper-fig13-routine-residual.svg --svg-levels paper/figures-
en/paper-fig14-routine-by-intensity.svg \
    --md paper/data/routine-coco-val2017.md
python3 paper/build_arxiv.py        # regenerates every figure and table, then builds this
document
```

4. Results

4.1 Non-geometric edits are practically invisible; geometric edits dominate

Table 1 and Figure 1 give the residual evasion rate per family and hash. Five of the seven families leave matching intact for all three hashes: under JPEG recompression down to quality 30, resize round-trips down to 15% of linear size and back, Gaussian blur to radius 3.5, additive noise to $\sigma = 35$ and brightness shifts of a quarter of the grey range, the residual rate is between 0.0% and 0.1% at every intensity, with Wilson upper bounds of 0.1% or less. The two geometric families are different in kind. PDQ, the deployed hash, fails to match 79.9% of center-cropped copies and 97.2% of rotated ones; pHash fails on 59.0% and 72.6%; Blockhash-GM on 53.2% and 64.6%. The ordering is the same on both families: the deployed hash is the most brittle to geometry and the most robust to everything else, which is the expected consequence of its aggressive downsampling and low-frequency selection, and the non-DCT hash is the most tolerant of geometry.

Table 1. Residual evasion rate by hash and transformation family on COCO val2017: share of edited images *not* matched to their own original at the threshold (PDQ 31/256; pHash 8/64; Blockhash-GM 31/256), pooled over four intensities, n = 20,000 image-intensity pairs per cell, 5,000 images per intensity, Wilson 95% interval in parentheses. Generated from `paper/data/routine-coco-val2017.json`.

Transformation family	Intensities	PDQ (Meta, 256 bit)	pHash (64 bit)	Blockhash-GM (256 bit)
Blur	radius 0.5 / 1 / 2 / 3.5	0.0% (0.0% to 0.0%)	0.0% (0.0% to 0.0%)	0.0% (0.0% to 0.0%)
Brightness	+10 / −10 / +25 / −25	0.0% (0.0% to 0.0%)	0.0% (0.0% to 0.1%)	0.1% (0.0% to 0.1%)
Center crop	keep 95 / 85 / 70 / 50 % of area	79.9% (79.3% to 80.4%)	59.0% (58.4% to 59.7%)	53.2% (52.5% to 53.9%)
Gaussian noise	σ 5 / 10 / 20 / 35	0.0% (0.0% to 0.1%)	0.0% (0.0% to 0.0%)	0.0% (0.0% to 0.1%)
JPEG compression	quality 90 / 70 / 50 / 30	0.0% (0.0% to 0.0%)	0.0% (0.0% to 0.0%)	0.1% (0.0% to 0.1%)
Resize round-trip	×0.8 / 0.5 / 0.3 / 0.15 and back	0.0% (0.0% to 0.1%)	0.0% (0.0% to 0.0%)	0.0% (0.0% to 0.0%)
Rotation	2° / 5° / 10° / 15°	97.2% (97.0% to 97.4%)	72.6% (71.9% to 73.2%)	64.6% (64.0% to 65.3%)

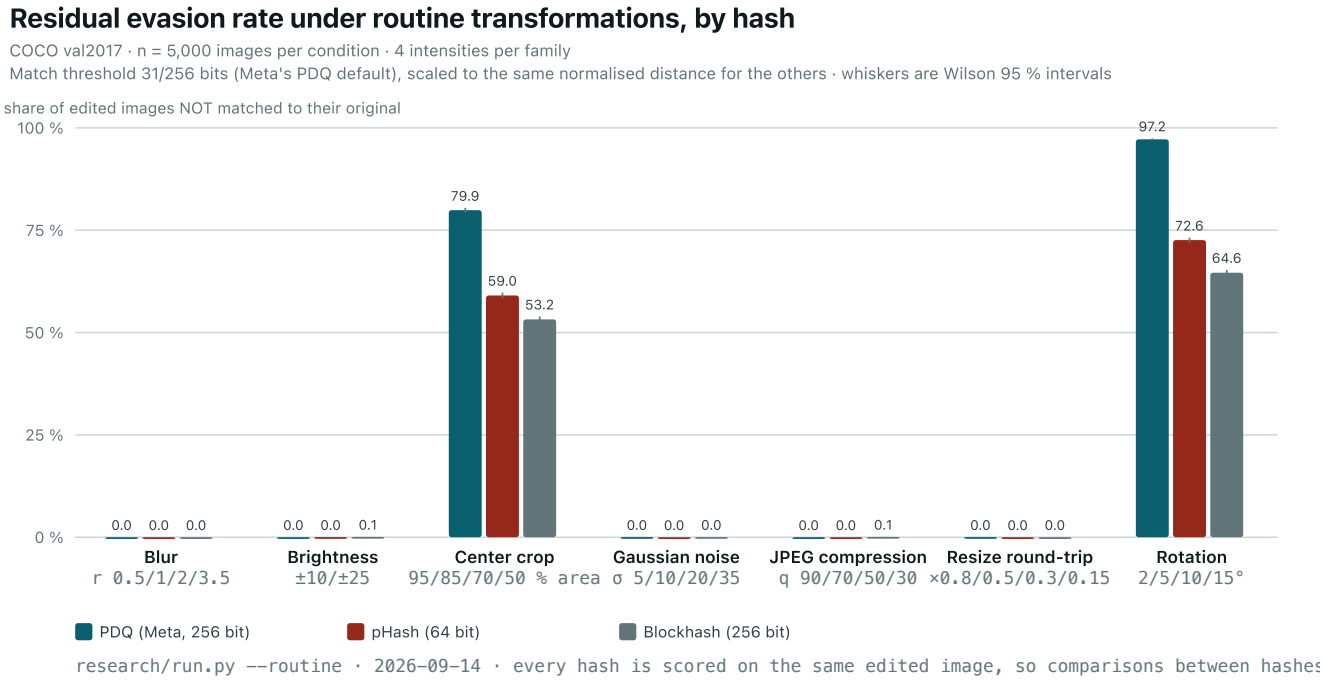


Figure 1. The same data as Table 1. Bars are residual evasion rates, whiskers are Wilson 95% intervals; at n = 20,000 the intervals on the geometric families are about half a percentage point wide and invisible at this scale. Every hash is scored on the same edited copy, so comparisons between bars within a family are paired. Rendered by `research/plot_routine.py`.

4.2 Where matching breaks

Figure 2 disaggregates the two geometric families by intensity. Center crop: keeping 95% of the area (removing a border of about 2.5% of each side, the kind of trim a screenshot or an app's default crop produces) already costs PDQ 19.8% of its matches; at 85% it loses 99.8%, and at 70% and 50% every copy. pHash and Blockhash-GM lose almost nothing at 95% (0.3% and 0.4%), 39.8% and 31.2% at 85%, and converge toward total failure at 70% and 50%. Rotation: at two degrees, a tilt small enough that a viewer may not notice it, PDQ fails to match 88.8% of copies; from five degrees it fails on every one. pHash loses 9.9% at two degrees and 81.0% at five; Blockhash-GM 2.0% and 60.5%; both are essentially defeated by ten degrees. The breaking points, in other words, are inside the range of edits that happen without intent: a few per cent of border, a couple of degrees of tilt.

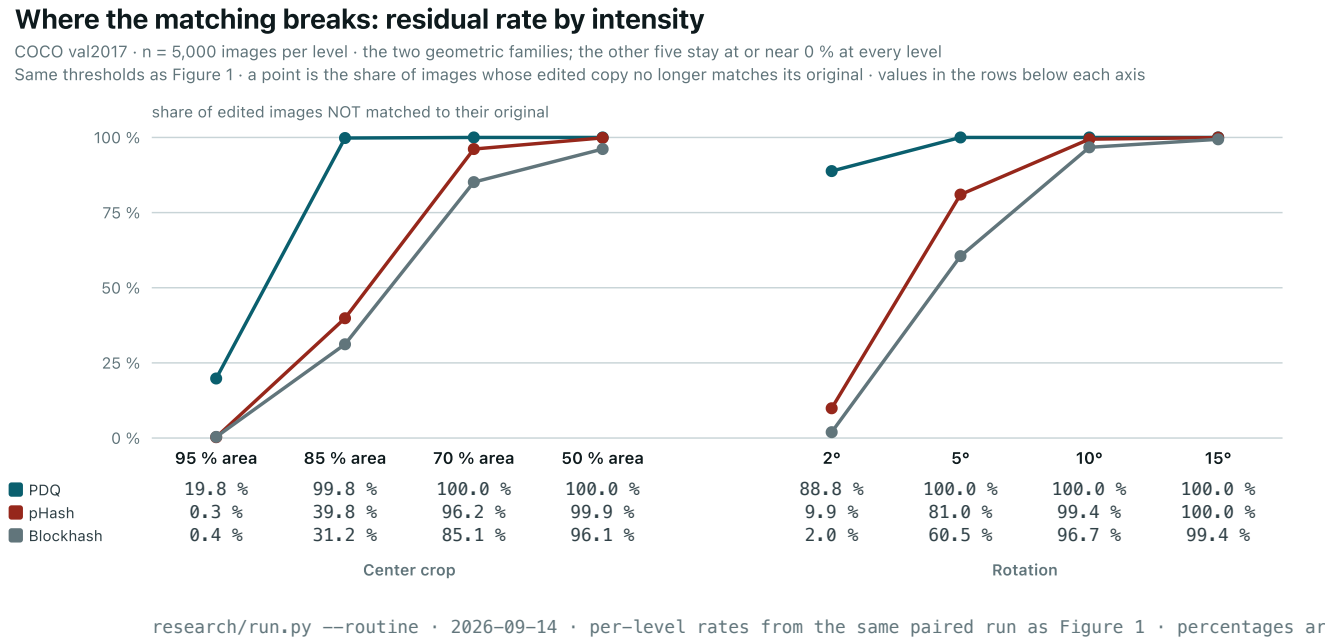


Figure 2. Residual evasion rate at each intensity for the two families that break matching, n = 5,000 images per point, same thresholds as Figure 1. The values are printed under each axis. The five non-geometric families are not drawn because they sit at or near 0% at every intensity (Appendix B gives all 84 cells). Rendered by `research/plot_routine.py` from the same results file.

4.3 Paired comparisons: a hash that fails differently

Table 2 gives the paired statistics. On center crop, Blockhash-GM's residual rate is 26.7 percentage points below PDQ's (bootstrap 95% interval −27.3 to −26.1 pp; 5,364 discordant pairs of which PDQ alone matched only 13; McNemar χ^2 = 5,310). On rotation the gap is 32.6 points (−33.0 to −32.1; χ^2 = 6,504). pHash sits between the two on both families (20.9 and 24.6 points better than PDQ; 5.8 and 7.9 points worse than Blockhash-GM), every interval excluding zero. The discordance with PDQ is almost entirely one-sided: of the 5,364 crop pairs and 6,522 rotation pairs on which PDQ and Blockhash-GM disagree, PDQ is the one that matched in only 13 and 4. Between pHash and Blockhash-GM the overlap is larger (pHash alone matched 877 of 2,920 discordant crop pairs and 564 of 2,716 rotation pairs), which is what two hashes from different families should look like: each recovers copies

the other loses. That asymmetry is the design content of the result. A second DCT descendant stacked on PDQ recovers part of the geometric loss; a hash from another family recovers more, and does so on different images.

The non-geometric rows carry the large-sample caution that any reader of a 20,000-pair comparison should keep in view. On JPEG recompression Blockhash-GM misses 13 copies that PDQ matches and PDQ misses none that Blockhash-GM matches, a difference of 0.1 percentage points with an exact binomial $p < 0.001$; on brightness, 13 against 2, 0.1 points, $p = 0.007$. Both are statistically significant and neither has any practical meaning: at this sample size, a difference of a dozen images is significant. We report the size of the effect and its interval alongside every p-value for that reason, and we read the non-geometric families as ties.

Table 2. Paired comparisons on identical image-transformation pairs. Difference is residual(B) minus residual(A) in percentage points, with a 95% bootstrap interval from 2,000 replicates resampled by image (seed 1). Discordant pairs are those on which exactly one hash matched; McNemar χ^2 with continuity correction, or the exact binomial test where fewer than 25 pairs are discordant. All three pairs are shown for the geometric families; PDQ versus Blockhash-GM for the rest. Generated from the same results file as Table 1.

Family	Pair (A vs. B)	Residual B – A (pp)	Bootstrap 95% CI (pp)	Discordant pairs (A only / B only)	McNemar χ^2	p
Center crop	PDQ vs. pHash	-20.9	-21.3 to -20.4	4,180 (4 / 4,176)	4,162.0	<0.001
Center crop	PDQ vs. Blockhash-GM	-26.7	-27.3 to -26.1	5,364 (13 / 5,351)	5,310.1	<0.001
Center crop	pHash vs. Blockhash-GM	-5.8	-6.4 to -5.3	2,920 (877 / 2,043)	464.8	<0.001
Rotation	PDQ vs. pHash	-24.6	-25.0 to -24.2	4,944 (9 / 4,935)	4,906.1	<0.001
Rotation	PDQ vs. Blockhash-GM	-32.6	-33.0 to -32.1	6,522 (4 / 6,518)	6,504.0	<0.001
Rotation	pHash vs. Blockhash-GM	-7.9	-8.5 to -7.4	2,716 (564 / 2,152)	927.3	<0.001
JPEG compression	PDQ vs. Blockhash-GM	+0.1	+0.0 to +0.1	13 (13 / 0)	exact binomial	<0.001
Brightness	PDQ vs. Blockhash-GM	+0.1	+0.0 to +0.1	15 (13 / 2)	exact binomial	0.007
Gaussian noise	PDQ vs. Blockhash-GM	+0.0	-0.0 to +0.1	12 (8 / 4)	exact binomial	0.388
Resize round-trip	PDQ vs. Blockhash-GM	-0.0	-0.0 to +0.0	6 (2 / 4)	exact binomial	0.688
Blur	PDQ vs. Blockhash-GM	-0.0	-0.0 to +0.0	3 (0 / 3)	exact binomial	0.250

4.4 What sample size the question needs

Table 3 answers the question a reviewer of any future comparison should ask first: could the study have seen the effect it claims? From PDQ's baseline, detecting a 5-point improvement on center crop at

$\alpha = 0.05$ and power 0.80 needs about 1,100 images per condition under the conservative independent-groups formula; a 30-point improvement needs about 40. On rotation, with its 97.2% baseline, 314 and 25. Studies of a few hundred images can therefore establish the large effects reported here and cannot resolve small ones, and a paired design needs fewer images than the table says, never more.

Table 3. Images per condition required to detect a given absolute reduction in residual rate from PDQ's baseline, two-proportion normal approximation, two-sided $\alpha = 0.05$, power 0.80, equal groups. The formula assumes independent groups and is conservative for the paired design used here.

Family	Baseline residual (PDQ)	5 pp	10 pp	15 pp	20 pp	30 pp
Center crop	79.9%	1,098	294	139	82	39
Rotation	97.2%	314	112	64	43	25

5. Discussion

5.1 What the measurement means for deployed interdiction

The deployed hash recognises a photograph through every non-geometric edit we tried and fails to recognise it after the two geometric edits that ordinary redistribution produces most often: a screenshot that trims a border, a phone that saves an image a couple of degrees off level, an app that crops to its own aspect ratio. In a system that enrols one hash per reported image and matches uploads against it, a copy cropped by 5% of its area is missed one time in five, and one cropped by 15% or tilted by two degrees is missed nine times in ten. These are the figures a platform, a regulator or a survivor should hold in mind when a removal duty is described as "known identical copies" of a reported image: the identical copy is caught; the copy that passed through one ordinary edit frequently is not.

Three remedies follow from the data, none of them new and each with a cost that our companion work prices. First, enrol more than one hash per image: the D_4 dihedral variants that PDQ's reference implementation can compute close the mirror and 90-degree holes (that is a separate measurement; the present protocol's rotations are small angles, which dihedral indexing does not address). Second, pair the deployed hash with one that fails differently: on the two families that matter, a block-mean hash recovers a quarter to a third of the copies PDQ loses, and recovers them on different images, which is the condition under which fusion buys anything; a second DCT descendant buys less. Third, address crop and rotation at the hash itself, as Steinebach, Berwanger and Liu (2022; 2023) proposed with segmentation-based hashing; we did not reproduce their method and make no claim about it beyond citing it as the right line of work. What the data rule out is the comfortable reading: that a single deployed hash with a single threshold, however well it survives compression, is robust to redistribution.

5.2 Governance after the white-box analysis and the inversion attacks

Two 2025 and 2026 results change what a hash match may be used to justify, and they bear on how robustness results like ours should be deployed. Deryck, Leblanc-Albarel and Preneel (2026) gave a mathematical description of PhotoDNA that is piece-wise linear and differentiable and constructed exact collisions and second preimages in seconds; the assurance that a hash cannot be forged is gone for the incumbent. Hawkes, Weinert, Almeida and Mehrnezhad (2025) showed that a low-budget generative inversion attack recovers a recognisable approximation of the original from PDQ and NeuralHash values, so that hash values uploaded in the clear by a survivor to a removal service "should be treated almost as carefully as the original images"; they recommend private set intersection so that the service never sees the raw hash, and advise users to weigh pre-emptive enrolment of material that may never be published. Our companion paper draws the architectural consequence, a routing system that holds hashes under a custody guarantee enforced in continuous integration and never auto-files any consequence from a match (Bravo and Castanon Acuna, 2026, section 4).

The present result adds a second-order governance point. The natural response to the geometric brittleness in Table 1 is to widen the net: raise the threshold, enrol more variants, add more hashes. Each of those raises the false-positive rate against the enrolled corpus, and after the inversion attacks every enrolled hash is itself a liability for the person who enrolled it. Robustness, false positives and hash custody are one trade, and a programme that reports only the first has not described the system. The honest statement of our data is therefore conditional: these are the recall losses at the deployed operating point; moving the operating point to recover them has a price in the other two quantities that a separate measurement (the inter-image census our companion work describes) is needed to pay.

5.3 The last two weeks: frontier and open-weight generators, and the fully synthetic gap

Everything above concerns an image that exists, was enrolled, and was then edited. Hash interdiction cannot, by construction, reach an image that was generated rather than copied: there is no source to enrol and no hash in any database to consume. Our companion paper states that limit as permanent (Bravo and Castanon Acuna, 2026, section 3.1, limitation L1). What the releases of the last two weeks of October and the first week of November 2026 change is not the limit but its reach, in three ways that primary sources let us date. Figure 3 and Table 4 list every release we rely on, each checked against its model card, repository or the developer's own announcement on 8 November 2026; the dated JSON behind them is committed with its sources.

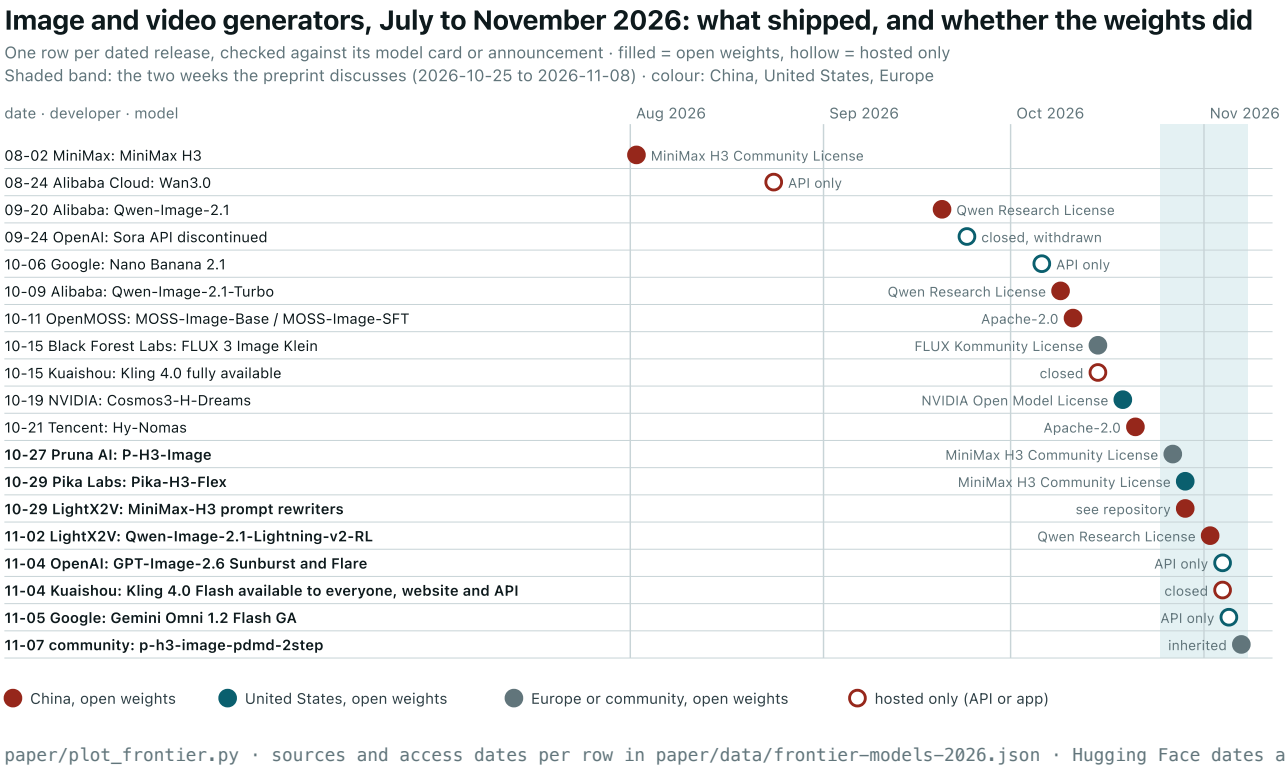


Figure 3. Image and video generator releases, July to November 2026, one row per dated event, from the developers' model cards and announcements. Filled marks are open weights (downloadable, runnable offline); hollow marks are hosted only. Bold rows fall inside the two weeks discussed in the text. Hugging Face dates are repository creation dates; X post dates are decoded from the status identifier. Rendered by `paper/plot_frontier.py` from `paper/data/frontier-models-2026.json`.

First, the hosted frontier moved, and kept its controls. OpenAI released GPT-Image-2.6 Sunburst and Flare to its API on 4 November 2026, with ChatGPT Images 2.6, adding SVG and animated-GIF generation and markup-guided editing; it had discontinued the Sora video API on 24 September. Google made Nano Banana 2.1 generally available on 6 October and Gemini Omni 1.2 Flash, its conversational video model with up to 30 seconds of extended output and six reference images, on 5 November; Google's image-generation documentation states that outputs carry a SynthID watermark. Kuaishou's Kling 4.0, a 30-second native video model that accepts up to 15 reference inputs, became fully available on 15 October, and Kling 4.0 Flash opened to everyone on the website and through the API on 4 November. Alibaba's Wan3.0 has been generally available through Alibaba Cloud since 24 August, hosted only. For hash interdiction these systems are the better case: the operator runs a concept filter before generation, moderates inputs, and can mark outputs, and a victim's complaint has an addressee. Nothing in a hosted release creates a source image to enrol, but the generation step at least passes through a point where a filter exists.

Second, the open-weight frontier moved faster, and in the direction of smaller hardware. Black Forest Labs released FLUX 3 Image Klein on 15 October: 5.6 billion parameters, native 4K, editing with up to ten reference images, a base checkpoint intended for fine-tuning with LoRA training supported from day one, and INT8 builds that the developer states run on 8 GB of consumer GPU memory and

produce a one-megapixel image in about four seconds on a current gaming card. Alibaba's Qwen-Image-2.1 (20 September; 7 billion parameters; generation and editing) was followed by its 8-step Turbo on 9 October and, on 2 November, by a third party's reinforcement-learning-refined 4-step distillation of the full transformer. MiniMax's H3, a 33-billion-parameter video model with native stereo audio whose weights were released on 2 August under a community licence, became the base of an ecosystem inside the window: Pruna AI distilled it into an 8-step, 8-megapixel image model on 27 October; Pika Labs released two LoRA adapters for 6-step video with audio on 29 October; LightX2V published prompt-rewriter models that replace the hosted preprocessing stage on 29 October; and a 2-step distillation of the Pruna image model appeared on 7 November. Tencent released Hy-Nomas, an interactive video world model, under Apache-2.0 on 21 October, and Fudan's OpenMOSS group released the MOSS-Image models under Apache-2.0 on 11 October. The pattern across both countries is the same: a frontier-adjacent generator, then within days a quantised or step-distilled derivative that halves the hardware and the time, then derivatives of the derivative.

Third, the only remaining control on open weights is the licence, and the licences say so. The MiniMax H3 Community License (release date 2 August 2026) defines an Applicable Territory that excludes the European Union, the United Kingdom, the Republic of Korea and the United States; prohibits use that exploits or harms minors and any circumvention of safety guardrails; and requires licensees to "add an AI-generation identifier to files produced". Pika Labs' adapters inherit that licence, so a United States company's open release is one that its own licence does not permit United States users to run; the Hugging Face repository recorded 516 likes in its first ten days regardless. The model card for H3 states that moderation of inputs and prompts applies to the hosted service; the weights themselves ship without it. Qwen-Image-2.1 tightened from the Apache-2.0 of its 2025 predecessor to a non-commercial research licence while the weights stayed public. The FLUX 3 Image Klein weights ship under Black Forest Labs' Kommunity License, and the model is built for fine-tuning on consumer hardware. None of this is a criticism of the developers, who are complying with the labelling regimes now in force (China's Measures for Labeling AI-Generated Synthetic Content since 1 September 2025; the EU AI Act's article 50 transparency obligations since 2 August 2026; the TAKE IT DOWN Act's removal duty since 19 May 2026). It is a description of where the control sits: in a text file that travels with the weights, enforceable after the fact against an identifiable party and against no one else.

What follows for interdiction by hash and for concept filters. For an image that was generated locally with an open-weight model, three things are true at once. There is no source image to enrol, so the victim-side hashing model of StopNCII and Take It Down has nothing to hash until the image has already been published, which is the failure the services exist to prevent. There is no durable mark: the provenance signals the hosted frontier applies (SynthID, C2PA manifests) are applied by the operator, and an offline generator has no operator; invisible watermarks are in any case removable by regeneration (Liu et al., 2025) and by the attacks catalogued in the WAVES benchmark (An et al., 2024), so their absence is no information. And there is no concept filter at generation: the guardrail that a hosted service runs on the prompt is a property of the service, and the licence clause that forbids bypassing it has no mechanism. What remains is the upload edge, where a platform can run a classifier on the generated image, and the after-the-fact hash: once a generated image has been reported and enrolled, hash matching applies to its redistribution exactly as it does to a photograph, with the

robustness profile measured in Section 4. The open-weight wave therefore shifts the burden from prevention to redistribution control, and makes the geometric brittleness in Table 1 the operative limit on the only tool that still applies. It also sharpens the case against any proposal to hash at the client: a client-side hash on a device that also runs a 5.6-billion-parameter image generator at four seconds per image is a hash on a device the user controls end to end. We state these as consequences of dated facts, not as predictions; the facts are in Table 4 and will be stale within weeks, which is itself part of the finding.

Table 4. Dated releases behind Section 5.3 and Figure 3, from 20 July 2026. Each row was checked against the source recorded in `paper/data/frontier-models-2026.json` on 8 November 2026. Bold rows fall inside the two weeks discussed. "Open weights" means the parameters are downloadable and the model runs without the developer's service; "API only" and "closed" mean the developer runs it.

Date	Developer	Model	Kind	Access	Licence
2026-08-02	MiniMax (CN)	MiniMax H3 (33B, video with native stereo audio)	video	open weights	MiniMax H3 Community License (release date 2026-08-02; excluded territories EU, UK, Republic of Korea, United States; separate authorization above USD 20M revenue; licensee must add an AI-generation identifier to outputs)
2026-08-24	Alibaba Cloud (CN)	Wan3.0 (up to 30 s native)	video	API only	hosted
2026-09-20	Alibaba (Qwen) (CN)	Qwen-Image-2.1 (7B generator, generation and editing)	image	open weights	Qwen Research License (non-commercial; release date 2026-09-20)
2026-09-24	OpenAI (US)	Sora API discontinued	video	closed, withdrawn	hosted
2026-10-06	Google (US)	Nano Banana 2.1 (gemini-nano-banana-2.1) GA	image	API only	hosted
2026-10-09	Alibaba (Qwen) (CN)	Qwen-Image-2.1-Turbo (8 steps)	image	open weights	Qwen Research License
2026-10-11	OpenMOSS (Fudan University) (CN)	MOSS-Image-Base / MOSS-Image-SFT	image	open weights	Apache-2.0
2026-10-15	Black Forest Labs (DE)	FLUX 3 Image Klein (5.6B, native 4K, editing, 10 references)	image	open weights	FLUX Kommunity License
2026-10-15	Kuaishou (Kling AI) (CN)	Kling 4.0 fully available (30 s native, up to 15 references)	video	closed	hosted
2026-10-19	NVIDIA (US)	Cosmos3-H-Dreams (action-conditioned surgical world model, image-to-video)	video	open weights	NVIDIA Open Model License

Date	Developer	Model	Kind	Access	Licence
2026-10-21	Tencent (Hunyuan) (CN)	Hy-Nomas (interactive video world model)	video	open weights	Apache-2.0
2026-10-27	**Pruna AI (FR)**	**P-H3-Image (image model distilled from MiniMax H3, 8 steps, 8 MP)**	**image**	**open weights**	**MiniMax H3 Community License (inherited)**
2026-10-29	**Pika Labs (US)**	**Pika-H3-Flex (two LoRA adapters on MiniMax H3, 6-step video with audio)**	**video**	**open weights**	**MiniMax H3 Community License (inherited; its Applicable Territory excludes the United States)**
2026-10-29	**LightX2V (CN)**	**MiniMax-H3 prompt rewriters (Qwen3.5-9B, Qwen3.8-27B)**	**video**	**open weights**	**see repository**
2026-11-02	**LightX2V (CN)**	**Qwen-Image-2.1-Lightning-v2-RL (full DiT, 4 steps, CFG 1.0)**	**image**	**open weights**	**Qwen Research License (inherited)**
2026-11-04	**OpenAI (US)**	**GPT-Image-2.6 Sunburst and Flare (API), ChatGPT Images 2.6**	**image**	**API only**	**hosted**
2026-11-04	**Kuaishou (Kling AI) (CN)**	**Kling 4.0 Flash available to everyone, website and API**	**video**	**closed**	**hosted**
2026-11-05	**Google (US)**	**Gemini Omni 1.2 Flash GA (video up to 15 s per turn, 30 s total, 6 references)**	**video**	**API only**	**hosted**
2026-11-07	**community (akhaliq) (n/a)**	**p-h3-image-pdmd-2step (2-step distillation of P-H3-Image)**	**image**	**open weights**	**inherited**

5.4 Relation to the companion paper and to the measurement programme

This note measures one quantity the companion paper names and does not measure: the robustness of the deployed primitive to routine edits. The companion paper's other measurement claims, the inter-image distribution and its false-positive tail, the error correlation across hashes, and the cost of dihedral indexing, are the subject of a separate census over 287,360 COCO 2017 photographs run in October 2026 (hash-once pipeline, every one of 4.1×10^{10} distinct pairs enumerated), reported by one of us as a research note with its code and results public (Castanon Acuna, 2026); nothing in this note depends on it, and its integration into the project's joint work is pending the review recorded in the repository. The two measurements are complementary by construction: this one asks how many true copies are lost at the operating point, the census asks how many unrelated images are confused at it.

6. Limitations

- **Benign photographs, not abuse imagery.** COCO val2017 contains everyday scenes. Robustness under a transformation is a property of the hash function and the transformation and is measurable on any photographs, but the images that reach removal services have their own distribution of sizes, compression histories and content, and the rates here should be read as a benchmark on photographs, not as a measurement on case material, which this project does not hold and will not hold.
- **Robustness, not the false-positive tail.** 5,000 images resolve residual rates to a fraction of a percentage point; they say nothing about inter-image collisions at 10^{-8}, which need on the order of a million distinct images and are measured separately (Section 5.4).
- **Non-adversarial by design.** No transformation here is optimised against the hash. Adversarial evasion is well studied (Section 2.3) and would make every geometric number worse and the non-geometric ones non-zero.
- **One operating point.** All rates are at the documented PDQ threshold of 31/256 and its scaled equivalents. A threshold sweep (say 0.08 to 0.15 normalised) would show how much of the geometric loss is recoverable at what false-positive cost; it is the natural next run and is not reported here.
- **The third hash is a variant.** Blockhash-GM thresholds at a global median and is not bit-identical to the Commons Machinery reference blockhash; its reference-exact counterpart differs in inter-image statistics. Its role here is a non-DCT robustness control and its numbers should not be quoted as the library's.
- **Implementation specifics.** Lanczos resampling, Pillow's JPEG encoder, bicubic rotation with black corners, and a per-image deterministic noise seed are choices; other choices change the numbers at the margins. All of them are fixed in code so that a different choice is a different, comparable experiment rather than an ambiguity.
- **Corpus geometry.** Most COCO images are 640 pixels on the long side and 480 on the short; PDQ's downsampling windows behave differently for sizes in different ranges, and a corpus of larger or smaller images could move the small-angle rotation numbers.
- **No PDQ quality filter.** The 9 images below the quality floor were kept. Removing them changes no rate by more than rounding; a deployment that discards them would see marginally lower residuals.
- **Still images only.** Video hashes (TMK+PDQF, vPDQ) and the video generators of Section 5.3 are outside the measured scope; the generators are discussed for what they imply about enrolment, not measured.
- **Section 5.3 is a dated snapshot.** Every release in Table 4 was verified on 8 November 2026 against a developer-controlled source; model cards and licences change, and the section should be read with its date.
- **Per-hash, not per-system.** Platforms may combine hashes, enrol variants, or use thresholds other than the documented default; our numbers describe the primitive at its documented operating

point, not any one platform's pipeline.

7. Conclusion

On 5,000 everyday photographs, the deployed perceptual hash survives every non-geometric edit we applied and loses most of its matches to the two geometric edits that ordinary redistribution produces: 79.9% of center-cropped copies and 97.2% of rotated ones, with breaking points at a few per cent of border and two degrees of tilt. A hash from a different mathematical family fails substantially less on exactly those edits and on different images, which is a measured argument for fusing dissimilar hashes. The protocol, the statistics and the figures regenerate from committed code and data in under half an hour on a laptop, and the corpus never enters the repository. The open-weight image and video generators released in the two weeks before this note was written do not change the structural limit of hash interdiction, that a generated image has no source to enrol; they move generation to hardware the user controls, where neither the operator's filter nor a durable mark exists, and so they make redistribution control, with the robustness profile measured here, the only tool that still applies to the output. Measuring that tool honestly, at its deployed operating point, on photographs anyone can download, is the smallest contribution we could make and the one we think is owed first.

Declarations

Author contributions (CRediT). *Lorena Bravo:* conceptualization (the framing of hash-based interdiction as a governance problem and of the residual population); methodology (the transformation set and the statistical plan of the September 2026 draft, which this note's protocol implements); writing, review and editing; project administration; institutional relations; the legal and policy analysis drawn on in Sections 1 and 5.2. *Cesar Castanon Acuna:* software (the measurement harness, the paired statistics module with its known-answer tests, the transformation implementations, the figure and table renderers, and the build pipeline); investigation (the run of record and its repetition); formal analysis; data curation (committed aggregate results and the dated, sourced release register behind Section 5.3); visualization; validation (reference verification against Crossref, arXiv, USENIX and the IACR ePrint archive on 8 November 2026); writing of the original draft of Sections 3, 4 and 5.3; the zero-custody architecture and verification infrastructure described in the companion paper. Both authors approved the version submitted and agree to be accountable for it. Author order follows the companion paper and the project's written authorship register.

Competing interests. L.B. is the founder and president of WomenCISO y MenCISO, I.A.P., the organisation with which C.C.A. is affiliated as Head of Engineering; readers should weigh the shared institutional relationship accordingly. L.B. is employed by Google in a role unrelated to this research; Google's image and video models are discussed in Section 5.3 as publicly documented releases, and the views expressed are the authors' own and not those of any employer. The authors have no financial interest in any hashing system, platform, or generative-model developer discussed.

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Ethics. The measurement uses a public benchmark of everyday photographs and aggregate statistics only. No abuse material, case data or personal data was accessed, held or produced; the harness refuses a corpus path inside the repository and refuses to run without an explicit benign-corpus confirmation, and the project's policy holds that any question requiring abuse material is answered inside an authorised holder's environment under their governance, not the project's. No human participants were involved.

Data and code availability. The harness (`research/`), the statistics module (`research/vdlab/stats.py`), the renderers (`research/plot_routine.py`, `paper/plot_frontier.py`), the build script (`paper/build_arxiv.py`), the aggregate results of record (`paper/data/routine-coco-val2017.json`) and the dated release register (`paper/data/frontier-models-2026.json`) are published in the public mirror of the project's paper and research directories, https://github.com/castacu0/routine-edits-measured (code MIT; manuscript, data and figures CC BY 4.0); the companion census code and results are at https://github.com/castacu0/pdq-census. The corpus is the public COCO 2017 validation set, obtained from the COCO distribution; no image, hash of an image, or image URL is in the repository.

AI-assistance disclosure. Portions of the harness, the figure code and the assembly of this manuscript were produced with large-language-model tooling operating under the repository's hard rules, with the authors as named human verifiers; every number in the text is checked at build time against the committed results file, and every reference and every release in Table 4 was verified by a human against the primary source named.

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Appendix A. Environment and run of record

Run of record: 14 September 2026, 18:42 local time, `research/run.py --routine`, seed 1, 5,000 images, 7 families × 4 intensities, 2,000 bootstrap replicates, 1,451.7 seconds on one core of an Apple M5 laptop. Repetition: 8 November 2026, same command, same corpus, in a virtual environment with Python 3.14.5, NumPy 2.5.1, Pillow 12.3.0 and pdqhash 0.2.8 (Meta's reference PDQ); it took 1,486 seconds and every aggregate (misses per family and intensity, discordant-pair counts, bootstrap intervals, sample-size table, images under the quality floor) was identical to the run of record, as expected for a deterministic protocol. Thresholds in bits: PDQ 31/256, pHash 8/64, Blockhash-GM 31/256. Images below Meta's PDQ quality floor (quality at or below 49): 9 of 5,000, kept.

Appendix B. Residual evasion rate at every intensity

Table B1. All 84 cells: residual evasion rate per family, intensity and hash, n = 5,000 images per cell, COCO val2017. Generated from the results file; the two geometric families are drawn in Figure 2.

Family	Level	PDQ (Meta, 256 bit)	pHash (64 bit)	Blockhash-GM (256 bit)
Blur	0.5	0.0%	0.0%	0.0%
Blur	1.0	0.0%	0.0%	0.0%
Blur	2.0	0.0%	0.0%	0.0%
Blur	3.5	0.1%	0.0%	0.0%
Brightness	10	0.0%	0.0%	0.1%
Brightness	-10	0.0%	0.0%	0.0%
Brightness	25	0.0%	0.0%	0.1%
Brightness	-25	0.0%	0.0%	0.1%
Center crop	0.95	19.8%	0.3%	0.4%
Center crop	0.85	99.8%	39.8%	31.2%
Center crop	0.7	100.0%	96.2%	85.1%
Center crop	0.5	100.0%	99.9%	96.1%
Gaussian noise	5	0.0%	0.0%	0.0%
Gaussian noise	10	0.0%	0.0%	0.0%
Gaussian noise	20	0.0%	0.0%	0.1%
Gaussian noise	35	0.1%	0.0%	0.1%
JPEG compression	90	0.0%	0.0%	0.0%
JPEG compression	70	0.0%	0.0%	0.0%
JPEG compression	50	0.0%	0.0%	0.0%
JPEG compression	30	0.0%	0.0%	0.2%
Resize round-trip	0.8	0.0%	0.0%	0.0%

Family	Level	PDQ (Meta, 256 bit)	pHash (64 bit)	Blockhash-GM (256 bit)
Resize round-trip	0.5	0.0%	0.0%	0.0%
Resize round-trip	0.3	0.0%	0.0%	0.0%
Resize round-trip	0.15	0.1%	0.0%	0.0%
Rotation	2	88.8%	9.9%	2.0%
Rotation	5	100.0%	81.0%	60.5%
Rotation	10	100.0%	99.4%	96.7%
Rotation	15	100.0%	100.0%	99.4%

Preprint v1 · 8 November 2026 · Bravo and Castanon Acuna · Corresponding: lorenabravo@berkeley.edu · cesar@castacu0.com · Code, committed data and figures: github.com/castacu0/routine-edits-measured · regenerate with `python3 paper/build_arxiv.py`